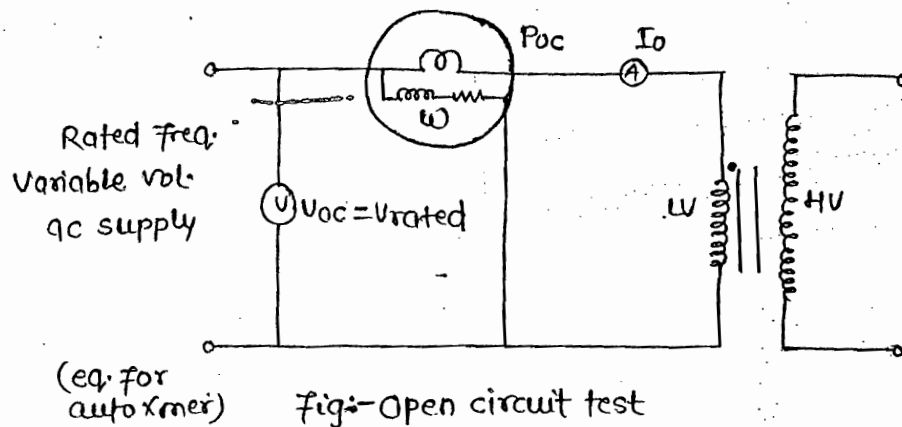


Open circuit Test & Short circuit Test



“To predict the performance without actual loading is the main purpose of performing above test.”

* The aim of these two test is to predict the performance of Xmer without actual loading.

OC Test →

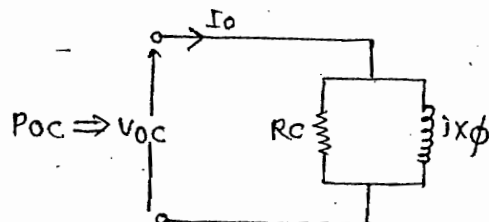
* This test is carried out at rated freq. & rated vol. to determine the core loss.

The core loss thus determined is then treated as constant in despite minor variation in freq. & voltage during actual operation.

* This test is carried out with the instruments placed on the LV side while the HV side is left open circuited.

This is because it is easier to arrange rated vol. supply at the low vol. level rather than at high vol. level.

* Also the instruments used are in economic in cost at LV level & it is safer to carry out the test.



eq. ckt under oc test

* Since the NL current is limited to 5% of FL value, the $I^2 R$ loss during OC test is ignored.

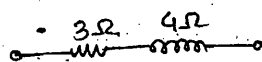
Also the $I^2 X$ impedance vol. drop at such low current is neglected.

* Since the NL PF is very low it is recommended that a low PF watt-meter is used in this test.

Que. → With the instruments connected on LV side OC readings are 10 kVA 450/120V, 50 Hz, 1- ϕ Xmer are 120V, 4.2A, 80W

Find the resistance & reactance of parallel eq. exciting ckt referred to HV side.

Soln. →



$$Y = \frac{1}{Z} = \frac{1}{3+4j} = \frac{1}{5\angle 53.13^\circ}$$

$$= 0.2\angle -53.13^\circ \text{ Siemens}$$

$$= 0.2(0.6 - 0.8j)$$

$$G - jB = 0.12 - 0.16j$$

$$G = 0.12, B = 0.16$$

$$P_{OC} = \frac{V_{OC}^2}{R_C}$$

$$R_C = \frac{(120)^2}{80} = 180\Omega$$

$$P_{OC} = V_{OC} I_0 \cos \phi_0$$

$$= 120 \times 4.2 \cos \phi_0$$

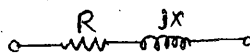
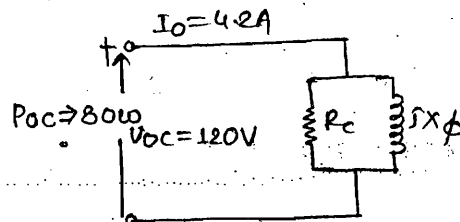
$$\phi_0 = 80.87^\circ$$

$$\tan \phi_0 = \frac{R_C}{X_\phi}$$

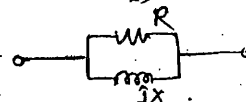
$$\tan(80.87^\circ) = \frac{R_C}{X_\phi}$$

$$X_\phi = \frac{R_C}{\tan(80.87^\circ)}$$

$$\boxed{X_\phi = 28.93\Omega}$$



$$\phi = \tan^{-1}\left(\frac{X}{R}\right)$$



$$\phi = \tan^{-1}\left(\frac{R}{X}\right)$$

$$\frac{1}{Z} = Y = \frac{1}{R} + \frac{1}{jX} = \frac{1}{R} - \frac{j}{X}$$

$$\tan \phi = \left(\frac{\frac{1}{X}}{\frac{1}{R}}\right)$$

$$\tan \phi = \frac{R}{X}$$

Ohmic value must be high on HV side & low on LV side.

HV side $\Rightarrow$

$$q = \frac{450}{120} = 3.75$$

$$R_c(HV) = (3.75)^2 \times 180$$

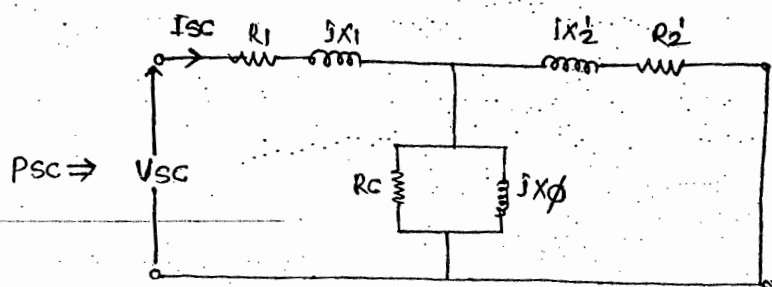
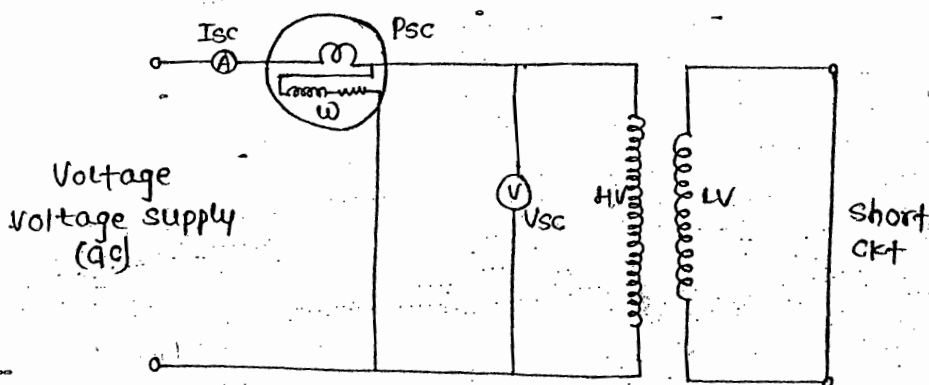
$$= 2.531 \text{ k}\Omega$$

$$X_\phi = (3.75)^2 \times 28.93$$

$$X_\phi = 406.83 \Omega$$

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Short circuit Test $\rightarrow$



* This test is carried out by rated current to determine FL cu loss.

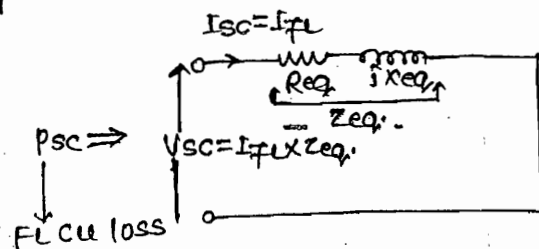
* Since the resistance of the wdg in Xmers is not affected much by the variation in power freq. band, this test need not be carried out strictly at rated freq.

Although it would be recommended to conduct the test at rated freq.

* This test is carried out with the instruments placed on the HV side while the LV side is short-circuited by a thick cond.

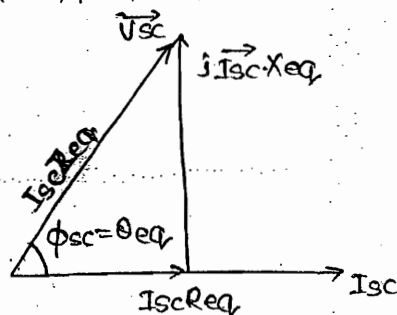
This is because the rated current on the HV side is less than on the LV side.

And therefore economically cheap instruments may be used in the test.



Since the applied vol. required to circulate FL current at SC would be limited to about 5% to 10% of the rated vol., the core loss at such reduced vol. is ignored.

Also the exciting current at such low vol. may be neglected.



$$Z_{eq} = \frac{V_{sc}}{I_{sc}} \cos^{-1} \frac{P_{sc}}{V_{sc} I_{sc}}$$

$$Z_{eq} = R_{eq} + jX_{eq}$$

Que. → With the ins. conncted in HV side the sc test readings for 50kVA 2200V/240V TF are 48V, 20.8A & 617W. Find the leakage impedance, effective resistance & leakage reactance referred to LV side.

Soln →

$$\phi = \cos^{-1} \frac{P_{sc}}{V_{sc} I_{sc}}$$

$$= \cos^{-1} \frac{617}{48 \times 20.8} = 51.83$$

$$Z_{eq} = 230 / 51.83$$

$$Z_{eq} = 1.421 + 1.808j \text{ } \Omega \text{ } 1.426 + j1.814$$

$$\boxed{Z_{eq} = 2.31 / 51.83^\circ \Omega}$$

LV side →

$$a = \frac{2400}{240} = 10$$

$$(R_{eq})_{LV} = \frac{1.426}{(10)^2} = 14.26m\Omega \quad (X_{eq})_{LV} = \frac{1.814}{(10)^2} = 18.14m\Omega$$

$$(Z_{eq})_{LV} = \frac{2.31 \Omega}{(10)^2} = 2.31m\Omega$$

Que. 7. A 2200/240V, 50Hz, 1 ϕ Xmer has exciting current of 0.6A has a core loss 361W, when its HV side is energised at rated voltage. Calc. the 2 components of exciting current.

(b) If a Xmer of part a supplies a load current of 60A at 0.8 PF lag on its LV side than cal. the 1 $^{\circ}$ current & its PF. Ignore leakage impedance drop.

Soln $\rightarrow$ (a) $I_0 = 0.6A$, $I_c = ?$, $I_\phi = ?$

$$P_{oc} = \frac{V_{oc}^2}{R_c}$$

$$P_{oc} = V_{oc} I_0 \cos \phi_0$$

$$361 = 2200 \times 0.6 \cos \phi_0$$

$$\cos \phi_0 = 74.13^\circ$$

$$I_0 = I_c + I_\phi$$

$$I_c = I_0 \cos \phi_0 = 0.164$$

$$I_0 = \frac{V_{oc}}{R_c} + \frac{V_{oc}}{X_\phi}$$

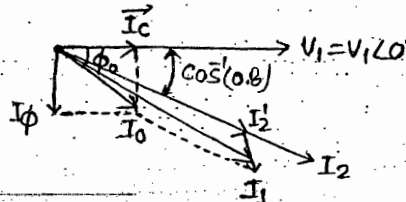
$$I_0 = 0.6 \angle -\cos^{-1} \frac{361}{2200 \times 240 \times 0.6}$$

Because always lag

$$I_0 = (0.1641 - j0.577) A$$

$$I_c$$

$$I_\phi$$



$$I_1 = I_2 + I_0$$

$$= \frac{I_2}{a} + I_0$$

$$= 6 \angle -\cos^{-1}(0.8) + (0.1641 - j0.577) j$$

$$= 6.488 \angle -40.08^\circ A$$

$$p/p \text{ PF} = \cos(40.08) \text{ lagging}$$

$$= 0.7651 \text{ lagging}$$

Interpretation of pu system →

$$R_{pu} = \frac{R_{eq}(\Omega)}{Z_{base}}$$

$$= \frac{R_{eq}(\Omega)}{V_{rated}/I_{rated}}$$

$$= \frac{I_{rated} \cdot R_{eq}(\Omega)}{V_{rated}} = \text{Full load resistance drop in \%}$$

Resistive drop = ohmic drop
= effective drop

$$= \frac{(I_{rated})^2 \cdot R_{eq}(\Omega)}{S_{rated}} = \text{Full load cu loss in per unit.}$$

$$X_{pu} = \frac{X_{eq}(\Omega)}{Z_{base}}$$

$$= \frac{X_{eq}(\Omega)}{V_{rated}/I_{rated}}$$

$$= \frac{I_{rated} \cdot X_{eq}(\Omega)}{V_{rated}} = \text{Full load reactive drop in \%}$$

$$= \frac{(I_{rated})^2 \cdot X_{eq}(\Omega)}{S_{rated}} = \text{Full load reactive loss in per unit.}$$

$$Z_{pu} = \frac{Z_{eq}(\Omega)}{Z_{base}}$$

$$= \frac{Z_{eq}(\Omega)}{V_{rated}/I_{rated}}$$

$$= \frac{I_{rated} \cdot Z_{eq}(\Omega)}{V_{rated}} = \text{Full load impedance drop in \%}$$

$$= \frac{(I_{rated})^2 \cdot Z_{eq}(\Omega)}{S_{rated}} = \text{Full load Apparent loss in per unit}$$